1 **(a)** 0.048 **[B1]**

(b) 380 **[B1]**

(c) $\frac{60 \times 4}{20 - 3}$ or $\frac{240}{17}$ **[M1]**

14 or 14.1 (from $\frac{240}{17}$) or 15 (if 17 is estimated as 20) **[A1]**

[Total Marks: 4]

2 **(a)** $\sqrt{100} \times 4^3$ or 10×64 $(= 640)$ **[M1]**

0.2×50 $(= 10)$ **[M1]**

$\frac{10 \times 64}{10} = 64$ **[A1]**

(b) Uses $a = 4$, $x = 0.05$, $b = 1$, $c = 0.8$ **[M1]**

$\frac{4 \times 0.05}{1 - 0.8}$ or $\frac{0.2}{0.2}$ **[M1]**

1 **[A1]**

[Total Marks: 6]

3 **(a)** **(i)** 47.54 **[B1]**

(ii) 48 **[B1]**

(iii) 48 **[B1]**

[Total Marks: 3]

4 $\frac{60 \times 3}{0.2}$ **[M1]**

$\frac{180}{0.2}$ **[M1]**

900 **[A1]**

[Total Marks: 3]

5 **(a)** $125 \times 74 = 9250$ **[M1]**

9300 **[A1]**

(b) $x^2 = 9250$ **[M1]**

$x = \sqrt{9250}$ (96.176...) **[M1]**

96.2 **[A1]**

[Total Marks: 5]

6 **(a)** $r \approx 4$, $h \approx 20$, density ≈ 8 **[M1]**

Estimated Mass $= 3 \times 4^2 \times 20 \times 8$ **[M1]**

7680 **[A1]**

(b) Volume $= \pi \times (4.15)^2 \times 19.8 = 1071.3\ldots$ **[M1]**

Mass $= 1071.3 \cdots \times 8.13 = 8710.19\ldots$ **[M1]**

8710 **[A1]**

[Total Marks: 6]